Group 2 - Markus Chambers, Vishal Green, Akshay Karthik, Trisha Bajpai, Ethan Bouryng

Requirements Analysis

1) Fully dressed use cases

- Akshay Karthik
  - UC1: About us page (donations link & blurb about making the world a better place)
    - Primary Actor: User
    - Stakeholders and Interests:
      - User wants to understand the mission of the application.
    - Preconditions: None
    - Success Guarantee:
      - User views the mission statement and has the option to donate or view app analytics.
    - Main Success Scenario:
      - User selects “About Us” from the navigation menu.
      - System displays a blurb regarding the app's mission to reduce food waste.
      - System displays a “Donate” link.
      - System displays a link to “Usage Analytics.”
    - Extensions:
      - If User is interested about their usage data → User is redirected to the graphs and analytics page which contains history of app use
    - Special Requirements: None
    - TDVL: None
  - UC2: History functionality (view graphs and analytics about history of app use)
    - Primary Actor: User
    - Stakeholders and Interests:
      - User wants to visualize their consumption habits and food waste over time.
    - Preconditions:
      - User has a history of adding/removing ingredients in the system.
    - Success Guarantee:
      - System renders visual data such as graphs representing app usage history.
    - Main Success Scenario:
      - User navigates to the “History and Analytics” section.
      - System retrieves historical data from the local database.
      - System generates and displays graphs (e.g., bar charts of most used ingredients).
    - Extensions: None
    - Special Requirements: None
    - TDVL: None
  - UC3: Reset password / set recovery email / 2FA

        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants to ensure their personal profile and inventory data remain private.
            - System needs to prevent unauthorized access.
        - Preconditions:
            - User is either logged in (to set 2FA) or at the login screen (to reset password).
        - Success Guarantee:
            - Security credentials are updated and verified.
        - Main Success Scenario:
            - User selects “Security Settings.”
            - User chooses to reset password, set a recovery email, or enable Two-Factor Authentication (2FA).
            - System sends a verification code to the User’s email/phone.
            - User enters the verification code.
            - System confirms the update and logs the security change.
        - Extensions: None
        - Special Requirements: None
        - TDVL: None
    - UC4: Offline Mode (save certain data for limited use when API is inaccessible for any reason)
        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants to check/update their fridge inventory in a place which has poor reception.
        - Preconditions:
            - The app has previously synced data while online.
        - Success Guarantee:
            - Cached inventory data is accessible without an internet connection.
        - Main Success Scenario:
            - System detects that a network connection is inaccessible.
            - System switches to “Offline Mode” by using local cache
            - User views their saved inventory list.
            - User makes an edit (e.g., marks an item as used).
            - System saves this change locally and flags it for a future sync up.
        - Extensions: None
        - Special Requirements: None
        - TDVL: None

- Markus
    - UC5: AI generator (meal plans and grocery list)
        - Primary Actor: System
        - Stakeholders and Interests:
            - System needs to generate weekly meal plans and grocery lists based on User inventory.
        - Preconditions:
            - User inventory exists.
        - Success Guarantee:
            - Meal plans and grocery lists are updated each week.
        - Main Success Scenario:
            - System retrieves inventory ingredients.
            - System selects recipes using those ingredients.
            - System generates a weekly meal plan with corresponding grocery lists.
            - System saves the meal plan and grocery lists and displays it to the User in designated area.
            - User can choose whether to keep or generate new meal plans.
        - Extensions:
            - If User chose to keep the meal plan → “Eat Meal” reminders are sent out during designated meal times.
        - Special Requirements: None
        - TDVL: None
    - UC6: “Eat Meal” Reminders
        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants reminders to maintain a consistent eating schedule.
        - Preconditions:
            - Inventory and recipe database available.
        - Success Guarantee:
            - User is reminded to eat via a notification
        - Main Success Scenario:
            - User pre-selects multiple meals/recipes to eat for one day
            - System analyzes choices from user.
            - System categorizes recipes in designated area.
            - User requests “Eat Meal” reminder within application.
            - System generates notification at specific hours of the day
        - Extensions: none
        - Special Requirements:
            - User must have pre-selected recipes prior to “Eat meal” reminder request
        - TDVL: None

- UC7: Barcode scanner in lieu of manual ingredient input
    - Primary Actor: User
    - Stakeholders and Interests:
        - User wants an easy way to add to inventory that doesn't include manual typing.
    - Preconditions:
        - User's device has a camera with granted permissions.
        - System is connected to a barcode lookup API.
    - Success Guarantee:
        - The item is identified and added to the inventory.
    - Main Success Scenario:
        - User selects "Scan Barcode".
        - The system activates the camera and lookup interface.
        - The system identifies the barcode and retrieves the product name and category.
        - System adds the item to the inventory.
    - Extensions: If barcode scanning was unsuccessful → User is redirected to add the ingredient manually.
    - Special Requirements: none
    - TDVL: None
- UC8: Contact us page
    - Primary Actor: User
    - Stakeholders and Interests:
        - User wants to resolve bugs or get help.
        - Support team needs clear information to resolve any issues with the system.
    - Preconditions:
        - User is logged into the application.
    - Success Guarantee:
        - The user's message is sent and a confirmation message is displayed.
    - Main Success Scenario:
        - User selects the "Contact Us" option.
        - The system displays a contact form (Subject, Category, Message).
        - User enters their query and clicks "Submit."
        - The system validates the text for any inappropriate content.
        - The system sends the message to the support team.
    - Extensions: none
    - Special Requirements: None
    - TDVL: None

- Ethan
    - UC9: Modes: Browse, Download, Upload
        - Primary Actor: User
        - Stakeholders and Interests:
            1. Users that would like to share their findings.
            2. Users that want suggestions / guidance from others.
        - Preconditions:
            1. User is signed in
        - Success Guarantee:
            1. User can navigate to Browse, Download, Upload options
            2. User can upload modes
            3. User can download modes
            4. User can create, manage, and modify downloaded modes
        - Main Success Scenario:
            1. User opens the modes library.
            2. System loads available modes.
            3. User browses mode descriptions.
        - Extensions:
            1. (Modify) & (Navigate) -> (Download/Upload)
        - Special Requirements:
            1. N/A
        - TDVL:
            1. SYSTEM api
            2. Mode must be valid
    - UC10: Set daily / monthly calorie intake level
        - Primary Actor: User
        - Stakeholders and Interests: User
        - Preconditions:
            1. User is signed in
            2. User is on profile / settings screen
        - Success Guarantee:
            1. Settings are saved
            2. Calorie intake level is fixed to new amt.
        - Main Success Scenario:
            1. User navigates to settings menu
            2. User adjusts parameters
            3. User saves settings
        - Extensions:
            1. N/A
        - Special Requirements:
        - TDVL:

            1. Valid calorie intake (daily) between 0 and 10,000
            2. Monthly calorie intake will set the dailyIntake field to (input / 30) to avoid variance between months
- UC11: Create and attach labels
    - Primary Actor: User
    - Stakeholders and Interests:
        1. Users that want to customize
    - Preconditions:
        1. User is signed in
        2. User is on
    - Success Guarantee:
        1. User-assigned labels display on Recipe Browser ingredients / recipes marked by user
    - Main Success Scenario:
        1. Label is successfully attached to ingredient / recipe
    - Extensions:
        1. Attach label to ingredient / Attach label to recipe
    - Special Requirements:
        1. N/A
    - TDVL
        1. Labels are Strings of max size 50 chars
- UC12: dashboard (automatic meal plans favoring usage of ingredients closer to expiry & site navigation)
    - Primary Actor: User
    - Stakeholders and Interests:
        1. User wants a convenient overview
        2. User wants a convenient and reasonable meal plan
    - Preconditions:
        1. User is signed in
        2. User is on dashboard screen
    - Success Guarantee: dashboard displays relevant info
    - Main Success Scenario:
        1. User opens dashboard from nav
    - Extensions:
        - N/A
    - Special Requirements:
        - N/A
    - TDVL
        - N/A

- Trisha
    - UC13: Manually Add Ingredients
        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants to update their digital inventory quickly.
            - System needs to maintain accurate inventory logs for effective recommendations.
        - Preconditions:
            - User has an active connection to the central ingredient database.
        - Success Guarantee: Ingredients are stored in inventory.
        - Main Success Scenario:
            - User selects “Add Ingredient”.
            - User enters ingredient information.
            - System validates the User’s ingredient input.
            - System saves the ingredient.
        - Extensions:
            - None
        - Special Requirements: None
        - TDVL: None
    - UC14: Shop Locator (Google Maps Functionality)
        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants to find the closest place to buy missing items quickly.
        - Preconditions:
            - Location services (GPS) are enabled on the device.
            - The user has identified missing ingredients from a recipe.
        - Success Guarantee: User is presented with directions to nearby stores stocked with any missing ingredients.
        - Main Success Scenario:
            - User views a recipe and selects “Find Missing Ingredients”.
            - System identifies which ingredients in the recipe are not in the User’s current inventory.
            - System requests User’s current GPS location and suggests nearby stores on a map.
            - User selects a store to view its hours of operation and its directions.
        - Extensions:
            - None
        - Special Requirements: None
        - TDVL: None
    - UC15: Cookbook management
        - Primary Actor: User
        - Stakeholders and Interests:

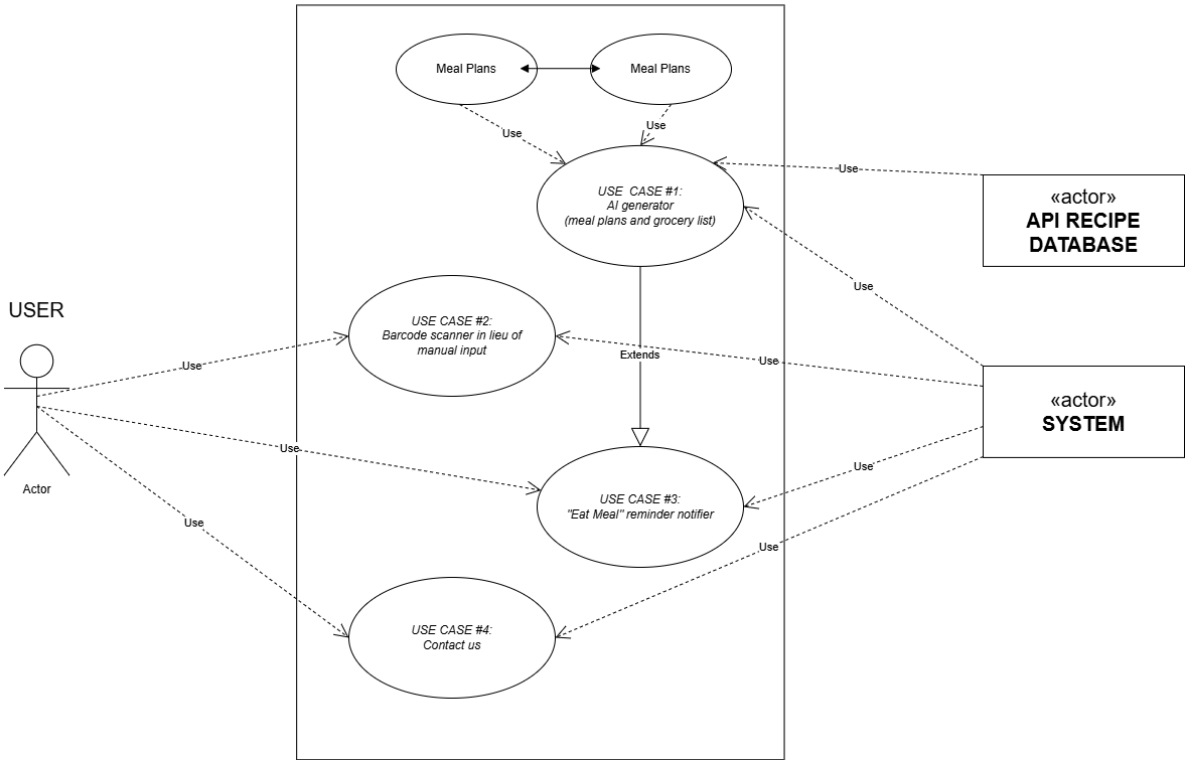
            - User wants to organize a digital cookbook and easily find meals based on what they own.
            - System needs to link recipes to the inventory database to show “missing” vs “available” items.
        - Preconditions:
            - Recipe database is accessible
        - Success Guarantee: The cookbook is updated with new or customized recipes.
        - Main Success Scenario:
            - User selects “Cookbook” from the main navigation menu.
            - User enters a recipe name or ingredient in the search bar.
            - System filters and displays matching recipes.
            - User selects “Edit” on an existing recipe.
            - User customizes recipe details and hits “Save”.
            - System saves the recipe to the User’s Digital Cookbook.
        - Extensions:
            - If some recipes have missing ingredients → User can use the shop locator to find nearby shops with the missing ingredients in stock.
        - Special Requirements: None
        - TDVL: None
    - UC16: Profile management
        - Primary Actor: User
        - Stakeholders and Interests:
            - User wants to make sure the app doesn’t suggest food that makes them sick (allergens) or don’t align with their dietary preferences.
            - System needs these constraints to filter all search results across the app.
        - Preconditions:
            - User is logged into their specific account.
        - Success Guarantee: User preferences and allergen constraints are saved.
        - Main Success Scenario:
            - User selects “Profile” from the navigation menu.
            - Use selects “Allergen Settings” and toggle specific items.
            - User sets "Dietary Preferences” (e.g. vegan, keto).
            - System saves the updated profile.
            - System applies these filters globally (e.g. Recipe search will now hide recipes containing selected allergens).
        - Extensions:
            - None
        - Special Requirements: None
        - TDVL: None

- Vishal
    - UC17: User Forum (similar to Reddit)
        - Primary Actor: User
        - Stakeholders and Interests:
            1. User has the ability to post thoughts and comments to a forum
        - Preconditions:
            1. User has an account
        - Success Guarantee: The system posts the comment to the forum
        - Main Success Scenario:
            1. User selects post topic
            2. User creates topic to post
        - Extensions:
            1. User does not have account → system prompts user to create an account
        - Special Requirements: none
        - TDVL: none
    - UC18: Official Help Guide
        - Primary Actor: User
        - Stakeholders and Interests:
            1. User has the ability to view and reference a help guide for the application
        - Preconditions:
            1. User has application
        - Success Guarantee: Help guide is accessible to user
        - Main Success Scenario:
            1. User selects view help guide
            2. Help guide is opened and viewable to user
        - Extensions:
            1. none
        - Special Requirements: none
        - TDVL: none
    - UC19: User account creation
        - Primary Actor: User
        - Stakeholders and Interests:
            1. User has the ability to create their own personal account
        - Preconditions:
            1. User has app installed
        - Success Guarantee: User can securely access their account.
        - Main Success Scenario:
            1. User creates account

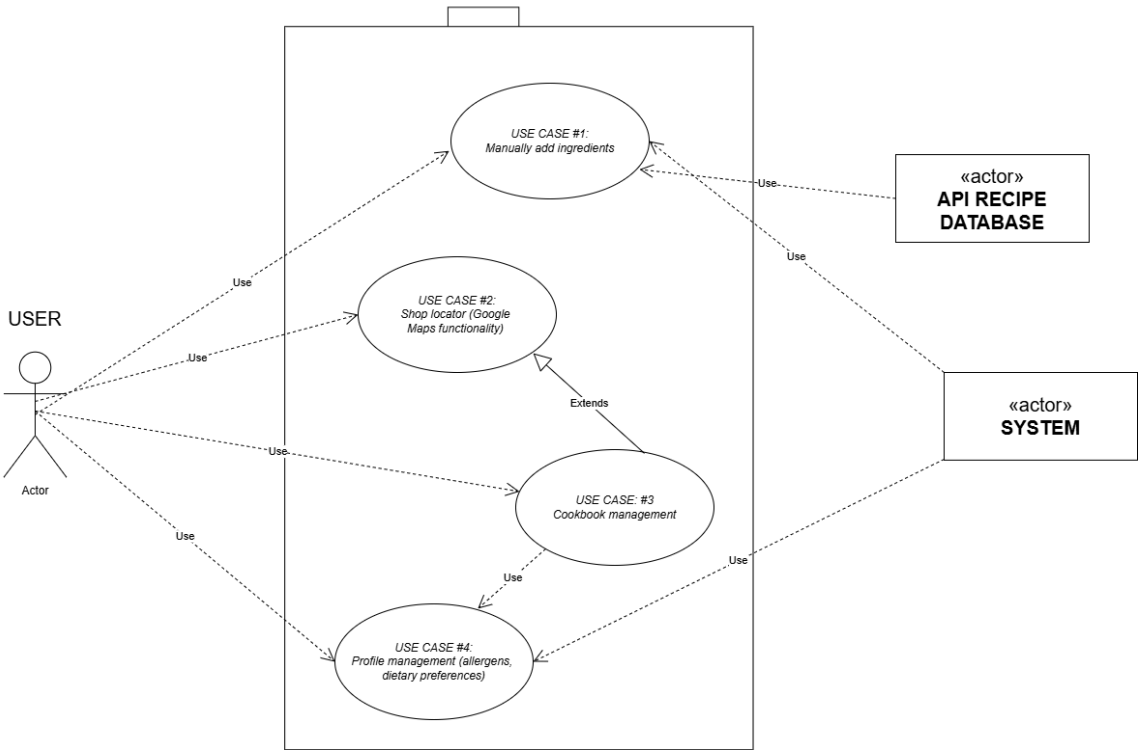
        2. System verifies credentials.
        3. User logs in successfully.
    - Extensions: user does not have account → system prompts user to create one
    - Special Requirements: none
    - TDVL: None
- UC20: Login/ Logout Functionality
    - Primary Actor: User
    - Stakeholders and Interests:
        1. Allows user to log into and log out of a previously created account
    - Preconditions:
        1. User must have an existing account
    - Success Guarantee: User's account is securely logged into or out of
    - Main Success Scenario:
        1. User enters account credentials
        2. User is securely logged into their account
        3. Upon selection of log out, user is securely logged out of their account
    - Extensions: if user does not have an account → system prompts user to create one
    - Special Requirements: none
    - TDVL: None

2) Use Case Diagrams

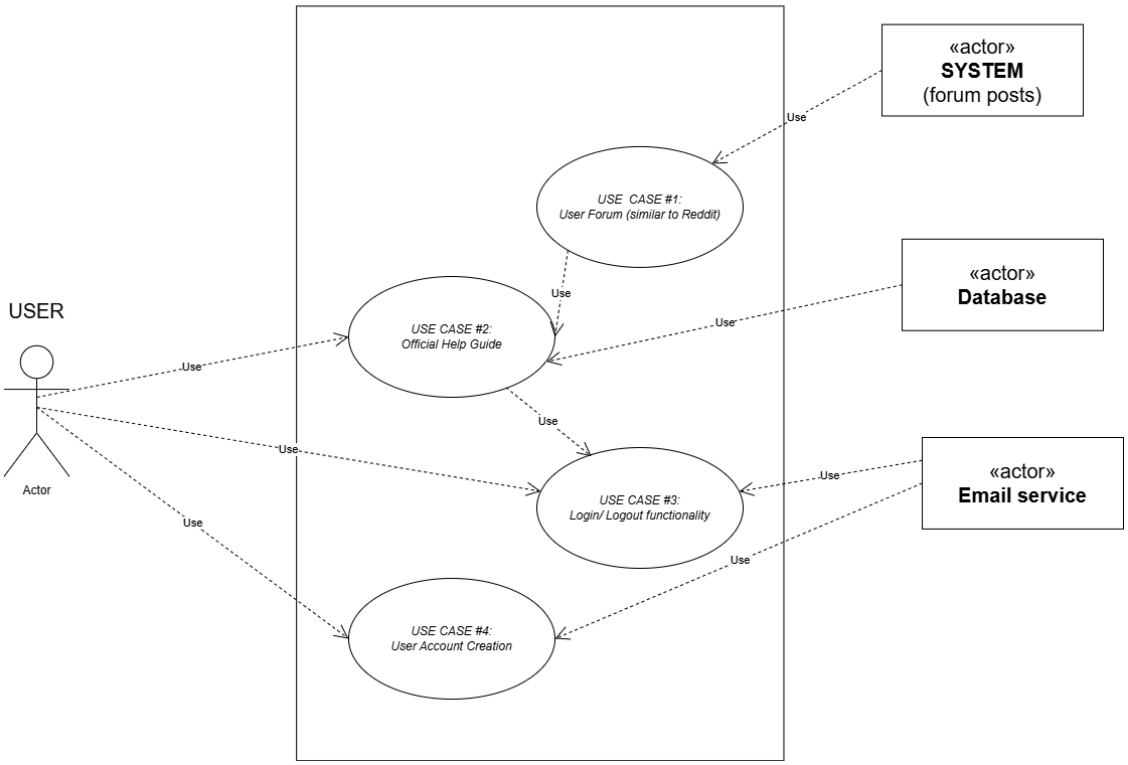
Markus Diagram



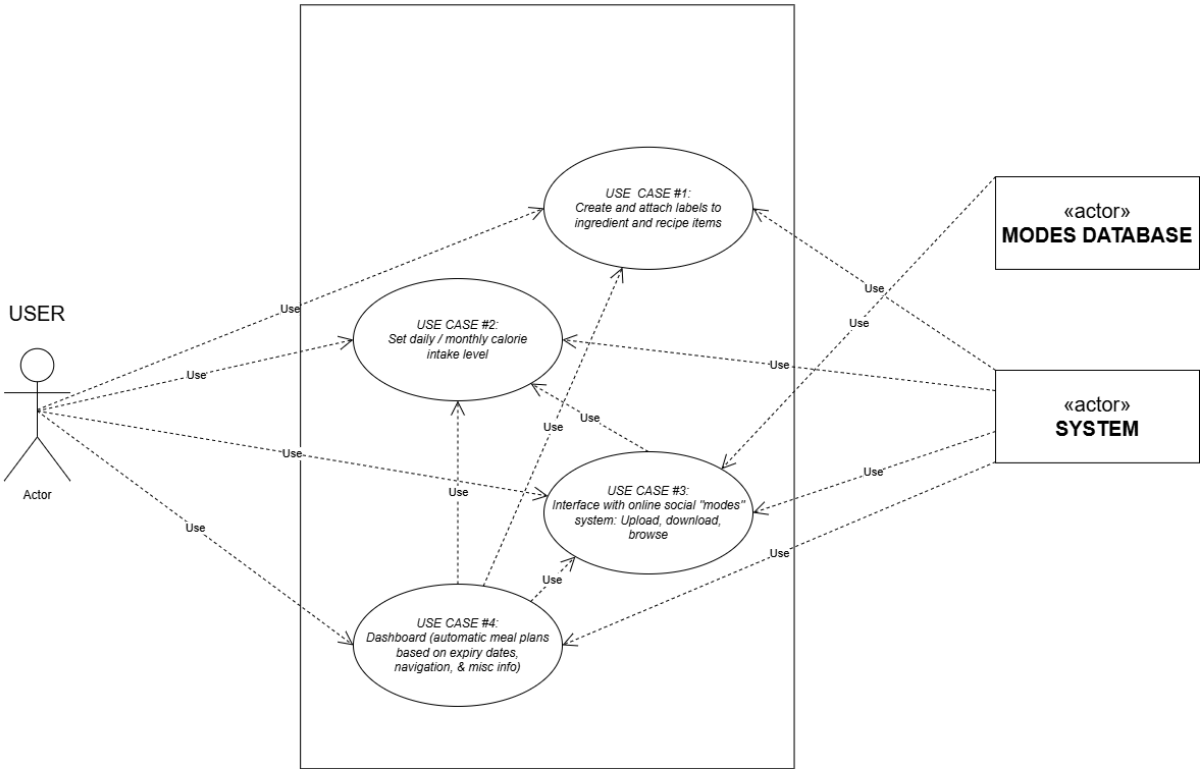
Trisha Diagram



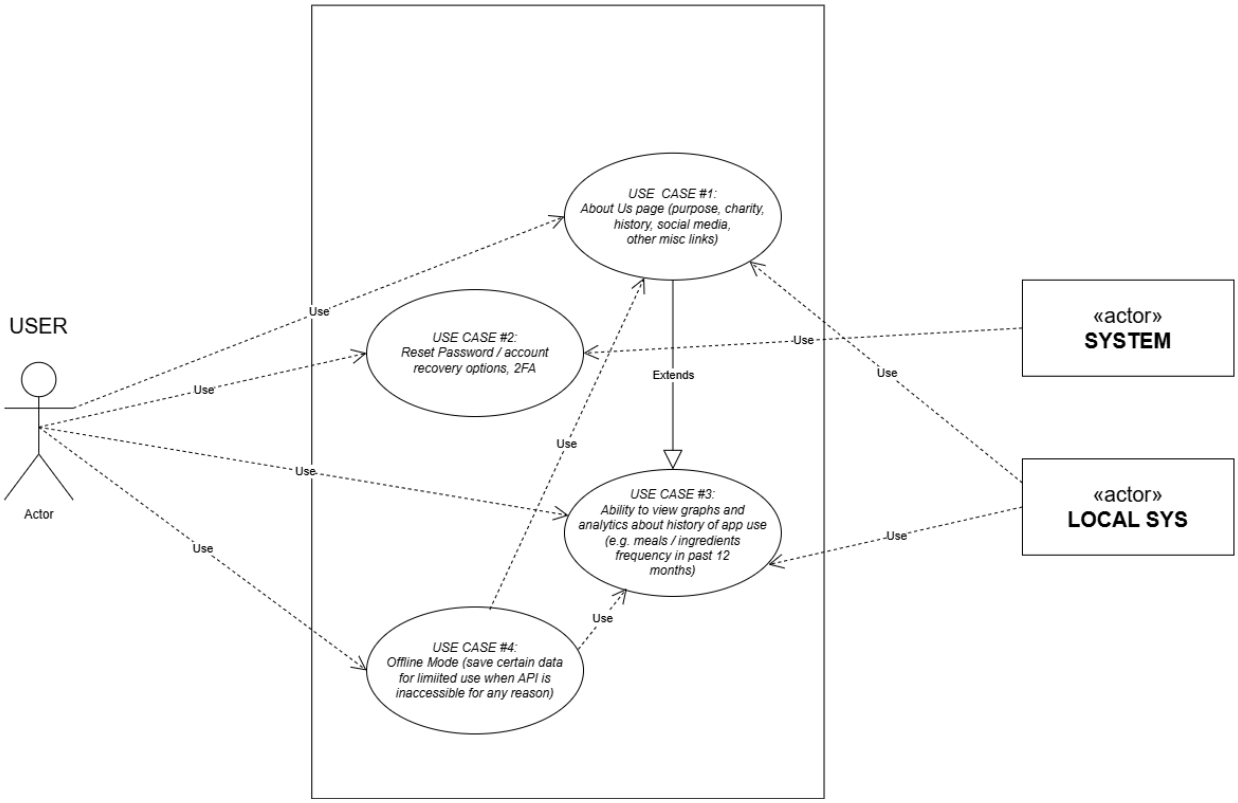
Vishal Diagram



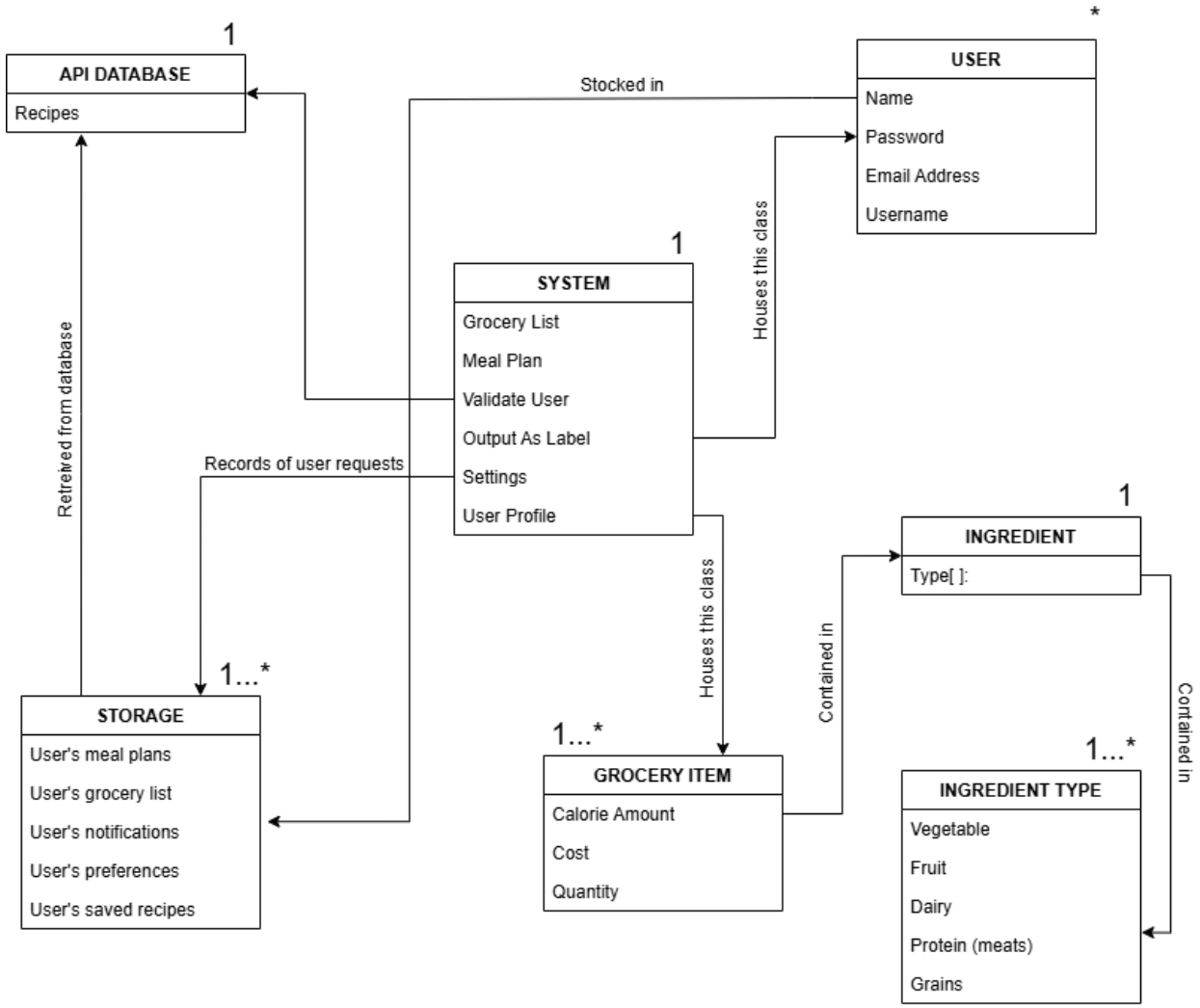
Ethan Diagram



Akshay Diagram



3) Conceptual Class Diagram



4) Supplementary Specifications

Our non-functional requirements include the following:

Markus: handle data entry sanitization

Ethan: The web interface will provide immediate visual feedback, such as a "Loading" GIF animation during background processing to inform the user that a request is in progress.

Trisha: All inventory items are loaded and displayed to the user within 10 seconds.

Akshay: The web interface will adhere to general accessibility standards by including a colorblind mode and ensuring screen reader compatibility.

Vishal: User accounts must load within 5 seconds and use secure authentication methods to protect user information.